***Out of Work**: Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators* 📈📉

Manish R. Kallu

DATA 510 Data Science Capstone · Summer 2026

WILLAMETTE UNIVERSITY

Abstract

Can the U.S. unemployment rate be predicted before it rises? *This project used public U.S. economic data to forecast the **national unemployment rate** 3, 6, and 12 months ahead. Results show that short-term forecasts are useful, while longer-term forecasts are less reliable.*

Introduction & Motivation

Problem & Question

Economic changes can quickly affect the U.S. unemployment rate, making early prediction valuable for planning. This study asks:

- *Can the **U.S. unemployment rate** be forecast **3, 6, and 12 months ahead** using public economic indicators?*
- *Which economic indicators have the **strongest historical relationship** with unemployment?*

Impact & Stakeholders

*The results can support **policymakers, economists, businesses, investors, and workforce agencies** in planning for future labor market changes.*

Background & Related Work

- *Broad economic indicators can improve macroeconomic forecasting **[1], [2]**.*
- *Consumer sentiment may provide information about future economic activity **[3]**.*
- ***This study** compares **3-, 6-, and 12-month forecasts** and examines **historical relationships between public economic indicators and unemployment.***

Data, Engineering & Ethics

Source & Scale:

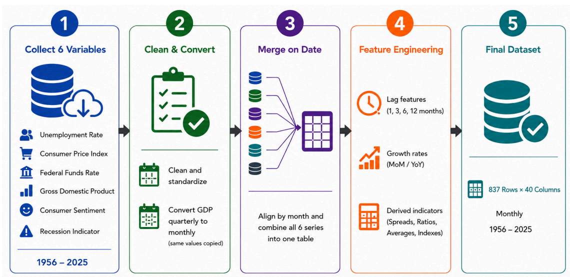
*Six public U.S. economic series produced **837 monthly records** from **April 1956 to December 2025**.*

Pipeline:

Download Raw Data → Clean → Merge by Date → Feature Engineering → Final Dataset

Ethics:

*The data are public and aggregated, with **no personal information.** Results should support planning, not replace human judgment.*



Methods & Evaluation

Approach:

*Compared **Linear Regression, Ridge Regression, Random Forest, Extra Trees, Gradient Boosting, and XGBoost** to evaluate **linear and nonlinear** forecasting across 3-, 6-, and 12-month horizons.*

Validation:

*Models were selected by **validation RMSE; tested using MAE, RMSE, and R^2.***

Evidence:

*Compared against **persistence** and **unemployment-only** baselines to verify performance gains.*

Results & Visualizations

Figure 1. Actual vs. predicted unemployment rate for the 3-month horizon using Ridge regression, Test set: n = 321 months

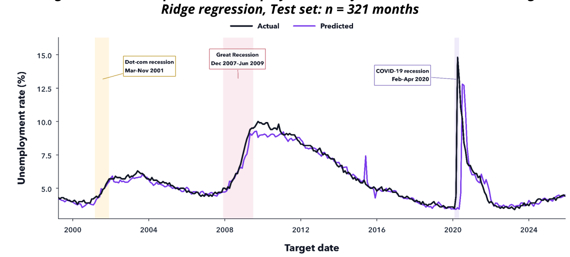
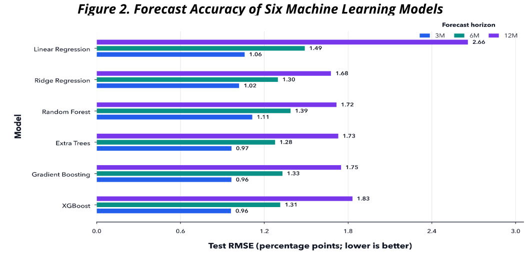


Figure 2. Forecast Accuracy of Six Machine Learning Models



Discussion, Limitations & Conclusions

Discussion:

- ***3-month forecasts** provided the strongest predictive performance; **6- and 12-month** accuracy declined.*

Limitations & future work:

- *The model uses **six national indicators** and does not capture **state- or industry-level** labor markets. Future work should include finer-grained data and evaluate additional forecasting methods.*

Conclusions:

- ***Short-term unemployment forecasting is practical. Consumer sentiment showed the strongest overall historical relationship with unemployment,** followed by **recession status.** These indicators can support earlier economic planning.*

Reproducibility & References

Repository: https://github.com/manishkallu01-wq/DATA510-Session-2 · Key dependencies: Python 3.13, pandas, NumPy, scikit-learn, matplotlib

References: [1] Stock & Watson (2002), Macroeconomic Forecasting Using Diffusion Indexes · [2] McCracken & Ng (2016), FRED-MD: A Monthly Database for Macroeconomic Research

[3] Barsky & Sims (2012), Consumer Confidence and Future Economic Activity